

*Best Security Solution that reduces cyber risk and increases
ship's stability*

IACS Rec. 190 (Jun. 2025) Guideline (Feat. How to fill out the vessel asset inventory)

CYBER SECURITY TEAM

Nov. 2025

*As ship network technology continues to evolve. Cyber Security is becoming a necessity rather than an option. **SeaNet** has standard business guidelines that comply with international regulations and continues to figure out domestic and international trends, providing customers with the best Cyber Security Solutions.*



Why “Vessel Asset Inventory” is necessary

According to 4.1.1.3 Requirement details of **IACS UR E26 Section 4.1 Identity 4.1.1. Vessel Asset Inventory**, which is mandatory on ships contracted for construction on or after 1 July 2024, **SeaNet** as a 3rd party System Integrator instead of the system integrator(Shipyard) requests you to fill out the vessel asset inventory to meet the requirement of IACS UR E26.

씨넷은 Shipyard 대행으로 IACS UR E26의 요구사항을 만족하기 위해, Vessel asset inventory 작성을 귀사에 요청합니다.

4.1.1.1 Requirement

An **inventory of hardware and software** (including application programs, operating systems, if any, firmware and other software components) of **the CBSs in the scope** of applicability of this UR and of the **networks connecting** such systems to each other and to other CBSs onboard or ashore shall be provided and **kept up to date** during the entire life of the ship.

이 UR의 적용 범위 내에 있는 **CBS의 하드웨어 및 소프트웨어**(애플리케이션 프로그램, 운영 체제, 펌웨어 및 기타 소프트웨어 구성 요소 포함)와 이러한 시스템을 **서로 연결하는 네트워크** 및 선상 또는 육상의 다른 CBS에 대한 목록은 선박의 전체 수명 동안 제공되고 **최신 상태로 유지**되어야 합니다.

4.1.1.3 Requirement details

The vessel asset inventory shall include at least the CBSs indicated in 1.3.2, if present onboard.

The inventory shall be kept updated during the entire life of the ship. **Software and hardware modifications** potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems shall be recorded in the inventory. If confidential information is included in the inventory (e.g. IP addresses, protocols, port numbers), special measures shall be adopted to limit the access to such information only to authorized people.

선박 자산 목록에는 1.3.2에 명시된 CBS가 탑재된 경우 최소 포함되어야 합니다.

재고는 선박의 전체 수명 동안 계속 업데이트되어야 합니다. 새로운 취약점을 도입하거나 시스템 간의 기능적 종속성 또는 연결을 수정할 가능성이 있는 **소프트웨어 및 하드웨어 수정 사항**은 재고에 기록되어야 합니다.

기밀 정보가 인벤토리에 포함된 경우(예: IP 주소, 프로토콜, 포트 번호), 이러한 정보에 대한 **접근을 권한이 있는 사람들만 제한**하는 특별 조치를 취해야 합니다.

E26	
E26 (cont)	
4. Requirements	
This section contains the requirements to be satisfied in order to achieve the primary goal defined in 3.1, organized according to the five functional elements identified in 3.2.	
The requirements shall be fulfilled by the stakeholders involved in the design, building and operation of the ship. Among them, the following stakeholders can be identified (see also section 2 for definitions):	
– Shipowner/Company – Systems integrator – Supplier – Classification Society	
Whilst the above requirements may be fulfilled by these stakeholders, for the purposes of this UR, responsibility to fulfil them will lie with the stakeholder who has contracted with the Classification Society.	
4.1 Identify	
The requirements for the 'Identify' functional element are aimed at identifying: on one side, the CBSs onboard, their interdependencies and the relevant information flows; on the other side, the key resources involved in their management, operation and governance, their roles and responsibilities.	
4.1.1 Vessel asset inventory	
4.1.1.1 Requirement	
An inventory of hardware and software (including application programs, operating systems, if any, firmware and other software components) of the CBSs in the scope of applicability of this UR and of the networks connecting such systems to each other and to other CBSs onboard or ashore shall be provided and kept up to date during the entire life of the ship.	
4.1.1.2 Rationale	
The inventory of CBSs onboard and relevant software used in OT systems, is essential for an effective management of cyber resilience of the ship, the main reason being that every CBS becomes a potential point of vulnerability. Cybercriminals can exploit unaccounted and out-of-date hardware and software to hack systems. Moreover, managing CBS assets enables Companies understand the criticality of each system to ship safety objectives.	
4.1.1.3 Requirement details	
The vessel asset inventory shall include at least the CBSs indicated in 1.3.2, if present onboard.	
The inventory shall be kept updated during the entire life of the ship. Software and hardware modifications potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems shall be recorded in the inventory.	
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What is CBS (Computer-Based System)

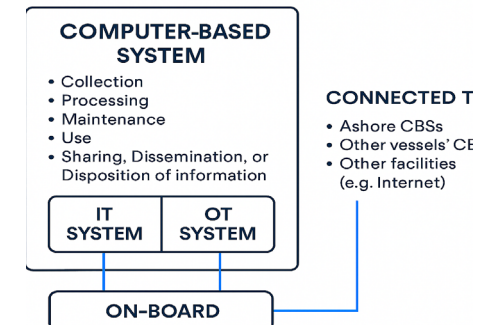
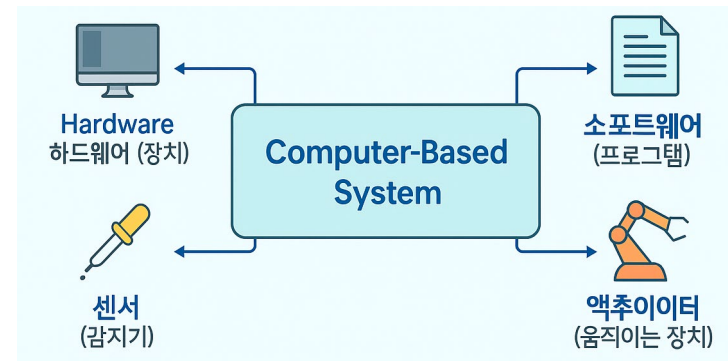
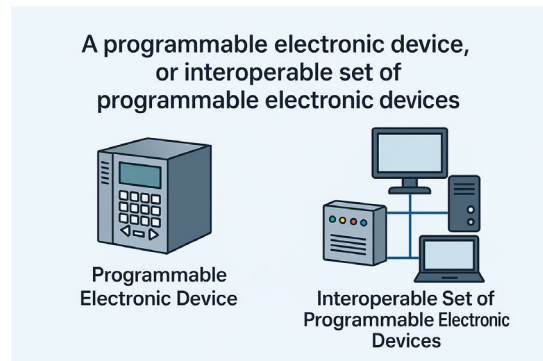
CBS Definition in IACS UR E22, 26, 27

A **programmable electronic device, or interoperable set of programmable electronic devices**, organized to achieve one or more specified purposes such as collection, processing, maintenance, use, sharing, dissemination, or disposition of information. CBS on-board include IT and OT systems. A CBS may be a combination of subsystems connected via network. On-board CBS may be connected directly or via public means of communications (e.g. Internet) to ashore CBSs, other vessels' CBS and/or other facilities.

CBS(Computer-Based System)는 정보를 수집, 처리, 유지관리, 활용, 공유, 전파 또는 폐기하는 등의 특정 목적을 달성하기 위해 구성된 **프로그래머블 전자 장치 또는 상호 운용 가능한 전자 장치들의 집합**을 의미합니다. 선박에 탑재된 CBS에는 IT 시스템(정보 기술)과 OT 시스템(운영 기술)이 모두 포함됩니다.

CBS는 여러 개의 서브시스템이 **네트워크를 통해 연결된 통합 시스템**일 수 있으며, **선박 내 CBS는 육상 CBS, 타 선박의 CBS, 또는 기타 시설과 직접 또는 공용 통신망(예: 인터넷)을 통해 연결될 수 있다.**

Term	Meaning	Example
Programmable Electronic Device	- An electronic machine that works by following instructions written in a program. - An electronic unit that executes functions based on software or firmware instructions stored within its computational architecture. 소프트웨어 또는 펌웨어를 통해 제어 로직을 설정할 수 있는 전자 제어 장치. PLC, IPC, RTU 등이 이에 포함됨.	- A device that turns a pump on or off based on temperature - A controller that manages a ship engine - An autopilot control unit - A smoke detector that reacts according to preset rules - PLCs, RTUs, HMIs, marine control units - Safety-related controllers or automation modules
Interoperable Set	- A group of electronic devices that can communicate with each other and work together. - A collection of programmable electronic devices that communicate through defined interfaces and protocols to operate as an integrated functional system. 서로 다른 전자장치가 통신을 통해 기능적으로 통합되어 작동할 수 있는 구조. 산업용 프로토콜(NMEA, Modbus, CAN 등)을 기반으로 함.	- A ship's integrated automation system (IAS): Sensors → controllers → displays → alarms → all connected - A fire detection system: Smoke detector → alarm panel → ventilation shutdown → fire door - A cooling control system: Temperature sensor → controller → pump → valve



Outline for Information in the Vessel Asset Inventory based on IACS Rec. 190 (Jun. 2025)

In order for the Vessel Asset Inventory to meet the intent of effective management of cyber resilience of the ship, the Vessel Asset Inventory should **include beneficial information about the items in the inventory**. So, we request you to complete the vessel asset inventory and for your easy understanding, we are providing this guideline with examples in the template .

Therefore, please check this guideline and fill out the vessel asset inventory, and then return to us.

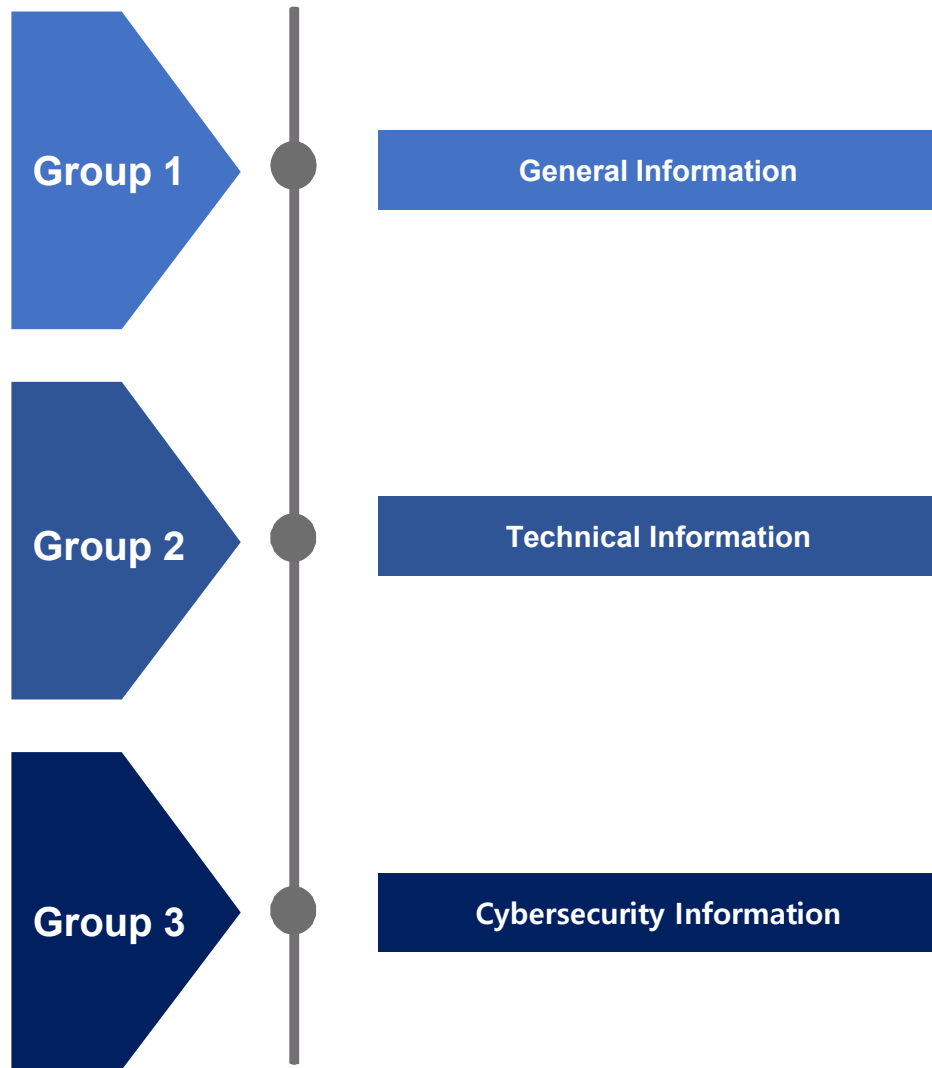
선박 자산 인벤토리가 선박의 사이버 복원력을 효과적으로 관리하려는 의도를 충족하기 위해서는 선박 자산 인벤토리는 다음과 같이 해야 합니다 인벤토리에 있는 항목에 대한 **유익한 정보를 포함합니다**. 이에, 선박 자산 인벤토리 작성을 요청 드리며, 쉽게 작성하기 위해 이 설명서를 예시 있는 양식과 함께 제공합니다. 설명들을 참조하여 작성하시어 회신 부탁드립니다.

Column	Subject	Description
A	Item Number	This column should indicate an item number for each row in the inventory. This is basic information to help identify each equipment item in the inventory
B	Ship functions and systems	This column should indicate the relevant ship function or system for the subject equipment.
C	System	This column should indicate the system name that will help identify the subject equipment.
D	Equipment	This column should indicate the equipment name that will help identify the subject equipment.
E	System Category	This column should indicate the System Category defined in UR E22 3.1. Most will be either Category II or Category III. Some will be Category I.
F	Brand/Manufacturer	This column should indicate the brand name or manufacturer name that will help identify the subject equipment.
G	Model / Type	This column should include the model identifier or type identifier that will help identify the subject equipment.
H	Unique Identifier	This column should include the system identification information required by paragraphs 4.2.2 and 7.1.1 of UR E22. This is for Category II and Category III equipment.
I	Operating System (OS)	This column should indicate the OS installed on the subject equipment (this includes physical and virtual machines).
J	Firmware	This column should indicate the firmware installed on the subject equipment, as well as the version and patch level .
K	Software and version	This column should indicate the application software and utility software for the subject equipment, as well as the version .
L	Location Onboard	This column should indicate the location of the subject equipment. It should also indicate the physical access restrictions that are provided.
M	Security Zone	This column should identify the security zone that the subject equipment belongs to.
N	Ship description of functionality/purpose	This column should indicate a short description of the functionality/purpose . This information is needed for hardware and software items .
O	Connection to components and systems with the scope	This column should indicate the connections to components and systems that are within the scope of E26 . The intent of this column is to identify the connections that exist with other components and systems that are within the scope of UR E26.

Outline for Information in the Vessel Asset Inventory based on IACS Rec. 190 (Jun. 2025)

Column	Subject	Description
P	Connections to Untrusted networks	This column should indicate the connections to components and systems that may appear to be not “in the scope of UR E26” . Please note that UR E26 considers any network outside the scope of applicability of UR E26 to be an “Untrusted network”.
Q	Physical interfaces	This column should indicate the physical interfaces for the subject equipment (such as network or serial interfaces).
R	Supported communication protocols	This column should indicate the relevant supported communication protocols for the subject equipment.
S	IP ranges/MAC address of nodes connected (if applicable)	This column should indicate the IP ranges and MAC addresses of the nodes connected (if applicable).
T	Certificate : Type Approval, etc., (if applicable)	This column should indicate information concerning any approval by a Class Society , such as a type approval, ship-specific approval, etc. (if applicable).
U	Excluded (if part of an excluded system in accordance with UR E26, Sec 6)	This column should remain blank for most equipment . This column should indicate an “X” if the equipment is part of a computer-based system (CBS) that has been excluded from meeting the requirements of UR E26.
V	Update log	This column should indicate an update log for the subject items. Software and hardware modifications potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems should be indicated in the update log column. Information should indicate when the modification was made, what the modification was and provide direction to Change Records.

Grouping Information in the Vessel Asset Inventory based on IACS Rec. 190 (Jun. 2025)



Column	Subject	Column	Subject
A	Item Number	F	Brand/Manufacturer
B	Ship functions and systems	G	Model / Type
C	System	H	Unique Identifier
D	Equipment	L	Location Onboard

Column	Subject	Column	Subject
I	Operating System (OS)	P	Connections to Untrusted networks
J	Firmware	Q	Physical interfaces
K	Software and version	R	Supported communication protocols
O	Connection to components and systems with the scope	S	IP ranges/MAC address of nodes connected (if applicable)

Column	Subject	Column	Subject
E	System Category	T	Certificate : Type Approval, etc., (if applicable)
M	Security Zone	U	Excluded (if part of an excluded system in accordance with UR E26, Sec 6)
N	Ship description of functionality /purpose	V	Update log

Group 1 – General Information

If you find the below table with examples, we think you can easily fill out the contents in Group 1. But regarding Column B & C, please find our explanation at the next page for you.
그룹 1 상에 요건들 하기 예제 확인하시어, 작성 부탁 드립니다. Column B 와 C에 관련하여, 다음 페이지에 설명 참고하시어 작성 부탁 드립니다.

Table 1. Examples							
Column A:	Column B:	Column C:	Column D:	Column F:	Column G:	Column H:	Column L:
Item Number	Ship functions and systems	System	Equipment	Brand/Manufacturer	Model/Type	Unique identifier	Location
1	Steering	Steering system	Steering control system	SCSUSUK	SCS ST01abc	SCS ST01abc	Wheelhouse
2	Propulsion	Main Propulsion System	Main Engine Control System (ECS) - Main Unit	ABC Machinery	MU-01-LNG	ECS-xx-xx-xx-xx-xA-Bx	Engine Room
3	Propulsion	Main Propulsion System	Main Engine Control System (ECS) - Wheelhouse Panel	XYZ Display	TFT10.4TP	ECS-xx-xx-xx-xx-xA-Bx	Wheelhouse
4	Propulsion	Main Propulsion System	Main Engine Control System (ECS) - Control Room Panel	XYZ Display	TFT10.4TP	ECS-xx-xx-xx-xx-xA-Bx	Control Room
5	Propulsion	Main Propulsion System	Ethernet Switch	DE networks	12mS5G	ECS-xx-xx-xx-xx-xA-Bx	Control Room
6	Propulsion	Main Propulsion System	Firewall	XYZ Firewall	Gate 09	ECS-xx-xx-xx-xx-xA-Bx	Control Room
7	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Main Unit	RCS Systems	RCS-001A	RCS-01-MU	Control Room
8	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Wheelhouse Panel	Disp RC AG	DP11EX	RCS-01-WH	Wheelhouse
9	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Control Room Panel	Disp RC AG	DP11EX	RCS-01-CR	Control Room
10	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Engine Room Panel	RCS Systems	RCS-001LC	RCS-01-LC	Engine Room
11	Propulsion	Main Propulsion System	Ethernet Switch	ESWX	100SW16	RCS-01-SW	Control Room
12	Watertight integrity and flooding detection	Water level detection and alarm system	Water Ingress Alarm System (WIAS) Alarm Unit	SEALEVELSWITCH	SEAAU-B	WIA-AU01	Wheelhouse
13	Watertight integrity and flooding detection	Water level detection and alarm system	Water Ingress Alarm System (WIAS) Sensor	SEALEVELSWITCH	SEASW-A	WIA-S01	Cargo Hold
14	Navigational systems required by statutory regulations	Navigation	Gyro Compass #1	North Star	NS-333	54293	Steering Stand
15	Navigational systems required by statutory regulations	Navigation	Navigation Workstation #1	Marine Company	NAV-WS-1234	13524	Wheelhouse Locker #1
16	Other - Boundary	Boundary (between systems)	Boundary Firewall	Firewall PDQ	Bound 101	23412	Bridge rack (Locked in rack)

Reference : Appendix 2 for a sample of a Vessel Asset Inventory in IACS Rec. 190 (Jun. 2025)

Group 1 – General Information (Column B)

Column B Regulation	Column B: (paragraph 4.1.1.3 in UR E26 requirement) Ship functions and systems This column should indicate the relevant ship function or system for the subject equipment. For most equipment, this column should indicate one of the items listed below. The list is from paragraph 1.3.2.a of UR E26 which describes the “systems in scope”. - Propulsion - Steering - Anchoring and mooring - Electrical power generation and distribution - Fire detection and extinguishing systems - Bilge and ballast systems, loading computer - Watertight integrity and flooding detection - Lighting (e.g. emergency lighting, low locations, navigation lights, etc.) - Any required safety system whose disruption or functional impairing may pose risks to ship operations (e.g. emergency shutdown system, cargo safety system, pressure vessel safety system, gas detection system, etc.) - Navigational systems required by statutory regulations - Internal and external communication systems required by class rules and statutory regulations The list from paragraph 1.3.2.b of UR E26 also includes any interface from CBSs in scope of this UR to any other systems. Examples of such cases are, but not limited to, the following: - passenger or visitor servicing and management systems - passenger-facing networks - administrative networks - crew welfare systems - any other systems connected to OT systems, either permanently or temporarily (e.g. during maintenance). In such cases “other” could be entered into this column along with a description. Such as “other – administrative network”
Actions taken	1. Please kindly check what ship function or systems(Propulsion ~ Other) your subject system or equipment is involved in. 2. Please fill out at the content in column B. For your easy understanding, please check the next page, which is system examples in Functions. 1. 귀사가 제공하는 시스템이나 장비가 어느 ship function 또는 systems에 포함되는지 확인하여 작성해주십시오. 2. Ship function 별, 시스템 예시 다음 페이지 작성하였으니 참조 해주십시오.

Group 1 – General Information (Column B)

System Examples in Functions

Functions	System
Propulsion	Main engine control and safety system
	Remote propulsion control system
	Integrated automation system (IAS)
Steering	Steering gear system
Anchoring and mooring	-
Electrical power generation and Distribution	Auxiliary engine control and safety system
	Emergency generator control and safety system
Fire detection and extinguishing systems	Fire detection and alarm system
Bilge and ballast systems, loading computer	Ballast water management system
	Remote tank gauging system
	Cargo control and monitoring system (loading system)
	Oil discharge monitoring and control system (ODME)
	Loading computer
	Inert Gas system
Watertight integrity and flooding detection	Water ingress detection system
Lighting	Emergency lighting
	Low locations lighting
	Navigation lights
Safety systems	Gas detection, Gas Sampling system
	Emergency shut-down system
	Sewage treatment system
	Incinerator
	cargo safety system
	Pressure vessel safety system

Functions	System
Navigational systems	Gyro Compass & Auto Pilot
	ECDIS
	Speed log
	Echo sounder
	Automatic identification system (AIS)
	GNSS receiver (DGPS/GLONASS Navigator)
	RADAR
	Bridge Navigation Watch Alarm System (BNWAS)
	Heading or track control system
	VDR
	Weather Fax
	Anemometer
	Magnetic Compass
	GPS Compass
	Satellite Speed Log
Internal and external communication systems	VHF radio
	MF/HF
	NAVTEX receiver
	Public address General alarm system (PAGA)
	Two-way voice communication
	VoIP Telephone
	UHF
Other	Smart Ship Platform
	Network Monitoring System

Group 1 – General Information (Column C & H)

Column C Regulation	Column C: (paragraph 4.1.1.3.1 in UR E26 requirement) System This column should indicate the system name that will help identify the subject equipment. The system name should indicate the most relevant description of the system and should complement the information already provided in Column B. Therefore, Column C does not need to repeat the information from Column B. In cases when “other” is entered in Column B, Column C should provide more specific information that will help identify the subject equipment.
Actions taken	1. Please kindly check what ship function or systems(Propulsion ~ Other) your subject system or equipment is involved in. 2. Please fill out the more specific information in column C. Please kindly find the example in Table 1. 1. 귀사가 제공하는 시스템이나 장비가 어느 ship function 또는 systems에 포함되는지 확인해주시요. 2. 보다 가장 관련 장비를 확인하는데 도움이 되는 특정 시스템 이름을 기재해주시요. Table 1 예시 참조 부탁드립니다.

Column H Regulation	Column H: (paragraphs 4.2.2 and 7.1.1 in UR E22) Unique identifier This column should include the system identification information required by paragraphs 4.2.2 and 7.1.1 of UR E22. This is for Category II and Category III equipment. This is not required for Category I, therefore in case the subject equipment is Category I, this column may indicate “not applicable”.
Actions taken	1. Please kindly mark your serial number of each physical equipment/device in your system. 2. Please kindly mark N/A if there is no physical equipment/device in your system, or your system has Category I. For your easy understanding, please kindly find Column E System Category in this guideline. 1. 귀사가 제공하는 시스템 상에 개별 물리적 장비/장치의 시리얼 번호를 기재해주시요. 2. 귀사의 시스템에 물리적 장비/장치가 없거나, 카테고리 I인 경우에는 N/A로 기재해주시요. 참고로 Column E System Category 에 Category 분류 확인하실 수 있습니다.

Group 1 – General Information (Column L)

Column L	Column L: (paragraph 4.2.4.3.1 in UR E26 requirement) Location Onboard
Regulation	This column should indicate the location of the subject equipment. It should also indicate the physical access restrictions that are provided (required for Category II and Category III).
Actions taken	1. Please kindly mark where your system is located at the vessel. 2. Please kindly check if you supply the rack(console, cabinet) with key. If you do, please mark the rack(console, cabinet) name. At this point, locking the rack(console, cabinet) is very important with the key. 3. Please kindly mark Yard if you don't know the installation location or you don't supply the rack(console, cabinet). For your easy understanding, please kindly find Column E System Category in this guideline. 1. 귀사가 제공하는 시스템이 어디에 설치되는지 기재해주시시오. 2. 잠금 장치를 갖춘 Rack(console, cabinet)이 공급되는 경우, 해당 Rack(console, cabinet) 이름을 기재해주시시오. 3. 시스템 설치 위치나 잠금 장치를 갖춘 Rack이 공급되지 않은 경우, Yard로 기재해주시시오. 참고로 Column E System Category 에 Category 분류 확인하실 수 있습니다.



Location Restriction



Space Restriction



The rack with above key is not okay

Group 2 – Technical Information

If you find the below table with examples and have your system [datasheet and manuals](#), you can fill out the contents in Group 2. But from Column I to S, please find our explanation at the next page for you. 그룹 2 상에 요건들 [하기 예제](#) 및 귀사의 [설명서 및 사양서](#) 확인 하시어, 작성 부탁 드립니다. Column I에서 S까지, 다음 페이지에 설명 참고하시어 작성 부탁 드립니다.

Table 2. Examples							
Column I:	Column J:	Column K:	Column O: (Recommended)	Column P: (Recommended)	Column Q:	Column R:	Column S: (Recommended)
Operating System (OS) / Version / Patch	Firmware / Version / Patch	Software and version	Connections in scope (UR E26)	Connections to Untrusted networks	Physical interfaces	Supported communication protocols	IP ranges / MAC addresses
Op Sys X v1.007	Industry PLC Corp's PLC firmware v2.023	Steer Straight v03	VDR	N/A	RS485 (2/2)	Modbus IP	XXX.XXX
Windows 10 22H2	-	MU-01-LNG v1.0	VDR, RCS, WIAS	Shore	LAN (2/2), USB (0/4), Serial (1/1)	TCP/IP	192.168.xx.x: xxx
Windows 10 22H2	-	TP-WH-LNG v1.0			LAN (1/2), USB (1/2)	TCP/IP	192.168.xx.x: xxx
Windows 10 22H2	-	TP-CR-LNG v1.10			LAN (1/2), USB (1/2)	TCP/IP	192.168.xx.x: xxx
NA	DE v1.0	NA			LAN (7/12)	TCP/IP	192.168.xx.x: xxx
NA	1214 Gate V1.0	Boundary Gate V1.0			LAN (4/5)	TCP/IP	192.168.xx.x: xxx
Windows 10 IoT LTSC 2019	-	RCS-001A v1.0	VDR, ECS, WIAS		LAN (2/2), USB (1/4), RS485 (2/2)	TCP/IP, Modbus	192.168.xx.x: xxx
Windows 10 IoT LTSC 2019	-	RCS-001B v1.0			LAN (1/2), USB (1/2)	TCP/IP	192.168.xx.x: xxx
Windows 10 IoT LTSC 2019	-	RCS-001C v1.0			LAN (1/2), USB (1/2)	TCP/IP	192.168.xx.x: xxx
Windows 10 IoT LTSC 2019	-	RCS-001L v1.0			LAN (1/2), USB (1/2)	TCP/IP	192.168.xx.x: xxx
NA	ESWX v1.0	NA			LAN (6/12), RS485 (1/2), RS422 (2/2)	TCP/IP, Modbus	192.168.xx.x: xxx
SEA WIA v1.0	Industry PLC Corp's PLC firmware v2.055	SEA Alarm v1.0	ECS, RCS, VDR		LAN (1/1), RS422 (2/2)	TCP/IP, Modbus	192.168.xx.x: xxx
-	-	-			RS422 (1/1)	Modbus	-
N/A	-	Gyro XX v1.0	Repeaters, ECDIS	None	USB, RJ45, serial	NMEA 0183	IP XXX.XXX
Windows 10 22H2	-	1) Application software v2.0; 2) malware protection v1.0; 3) backup and restore v1.0; 4) Nginx v27.0; 5) Vmware v15.0	ECDIS	Shore	USB *3, RJ45 *2, CD-ROM *1, RS232/485/422 *1	MODBUS-TCP (RJ45 2#); NMEA 0183 (RS485)	IP XXX.XXX
NA	-	Boundary v1.1	Navigation System	Shore	USB *1, RJ45 *8	HTTPS, SSH	IP XXX.XXX

Group 2 – Technical Information (Column I & J)

Column I Regulation	Column I: (paragraph 4.1.1.3.1 in UR E26 requirement) Operating System - (OS) Version and Patch level included This column should indicate the OS installed on the subject equipment (this includes physical and virtual machines). The name of the OS should be indicated (such as Linux, Windows, etc.) as well as the version and patch level.
Actions taken	1. Please kindly check if you have any OS(operating system) in your system or equipment. 2. If you have any OS in your system or equipment, please mark “OS name with OS version and patch level”. 3. If you don't have any OS in your system or equipment, please mark “N/A”. 1. 귀사의 시스템이나 장비가 OS(운영 시스템)을 가지고 있는지 확인하십시오. 2. 만약 OS 를 가지고 있다면, OS 이름(OS version 및 patch level)를 기재하십시오. 3. 만약 OS 를 가지고 있지 않다면, N/A로 기재하십시오.

Column J Regulation	Column J: (paragraph 4.1.1.3.1 in UR E26 requirement) Firmware -Version and Patch level included This column should indicate the firmware installed on the subject equipment, as well as the version and patch level. This includes firmware installed on embedded devices and implement the operation of software on hardware, such as RTOS (Freertos, ucos, ...), software installed on switches /routers, etc.
Actions taken	1. Please kindly check if you have any firmware in your system or equipment. 2. If you have any firmware in your system or equipment, please mark “firmware name with firmware version and patch level”. 3. If you don't have any firmware in your system or equipment, please mark “N/A”. 1. 귀사의 시스템이나 장비가 firmware을 가지고 있는지 확인하십시오. 2. 만약 firmware 를 가지고 있다면, firmware 이름(firmware version 및 patch level)를 기재하십시오. 3. 만약 firmware 를 가지고 있지 않다면, N/A로 기재하십시오.

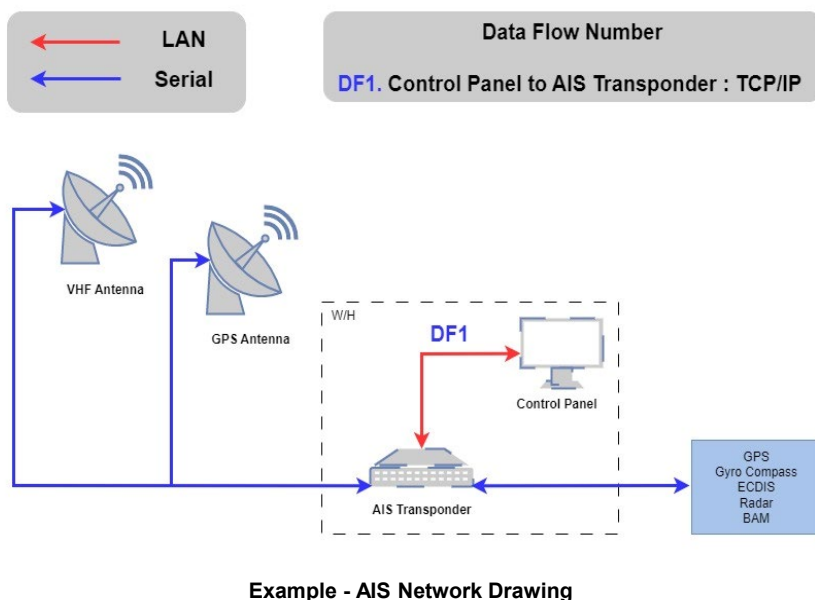
Group 2 – Technical Information (Column K)

Column K Regulation	Column K: (paragraph 4.1.1.3.2 in UR E26 requirement) Software and version This column should indicate the application software and utility software for the subject equipment, as well as the version. This includes programs that implement functions of onboard systems, and programs that are useful for files managing, system monitoring and protection, process and service management, backups and restore utilities.
Actions taken	1. Please kindly check if you have any application software and utility software in your system or equipment. 2. If you have any application and utility software in your system or equipment, please mark “application and utility software name with the version”. 3. If you don't have any application and utility software in your system or equipment, please mark “N/A”. For your easy understanding, please kindly find the below explanation about application and utility software 1. 귀사의 시스템이나 장비가 application & utility software을 가지고 있는지 확인하십시오. 2. 만약 해당 소프트웨어를 가지고 있다면, 해당 software 이름(version 포함)를 기재하십시오. 3. 만약 해당 소프트웨어를 가지고 있지 않다면, N/A로 기재하십시오. Application 및 utility software 대해 아래 설명 참조하십시오.

Software	Definition	Feature	Examples
Application	- Software that implements on the OS and performs actual "tasks/functions" "Functional programs" with clear purpose of use OS 위에서 돌아가며, 실제 “업무/기능”을 수행하는 소프트웨어 사용 목적이 분명한 “기능 프로그램”	- Requires functional version management - a layer that is frequently updated/replaced by people (engineers/vendors) 기능 단위 버전 관리 필요 사람(엔지니어/벤더)이 자주 업데이트/교체하는 층	Main Engine Control Application Ballast Control Application Water Ingress Alarm HMI program ECDIS, VDR viewer, Navigation Workstation App Loading Computer Software
Utility	Although it does not perform "operational purpose functions" directly like the Application does, Auxiliary software to assist with system operation and maintenance Application처럼 “운영 목적 기능”을 직접 수행하지는 않지만, 시스템 운영·유지보수를 도와주는 보조 소프트웨어	Not an Application, but Operations Required Application은 아니지만 운영 필수	Backup Utility v1.0 Restore Utility v1.0 Malware Protection v1.0 Nginx, VMware System Monitoring Tool SNMP Tools File Manager

Group 2 – Technical Information (Column O)

Column O Regulation	Column O: (recommended) Connections to components and systems within the scope This column should indicate the connections to components and systems that are within the scope of E26. The intent of this column is to identify the connections that exist with other components and systems that are within the scope of UR E26. The intent of this column is also to determine that all connections have been identified and whether they are within scope or outside scope (Please also see the description for Column P concerning connections to an “Untrusted network”).
Actions taken	1. Please check if your system/device has any connections to other components or systems. 2. Please kindly mark the communication type such as LAN and serial for each connected system/device if your system/device has any connections to others. 3. Please mark N/A if your system/device doesn't have any connections to other components or systems. 1. 귀사의 시스템이나 장치가 타 구성품 또는 시스템과 연결을 가지는 지 확인 부탁드립니다. 2. 귀사의 시스템이나 장치가 타 시스템이나 장치에 연결이 있다면, 통신 종류(Lan 통신 or Serial 통신 등)에 대해 기재해주십시오. 3. 타 시스템이나 장치에 연결이 없다면, N/A로 기재해주십시오. 예: AMS(RS485), VDR(RS422), Smart Ship Solution(LAN)



For example, let's check AIS system.

AIS system two components such as AIS transponder and control panel.

As you can see the drawing, there is the connection between AIS transponder and control panel.

At this moment, you need to check what communication type between AIS transponder and control panel is. It is Lan. You can find that AIS transponder has the connections to GPS, Gyro compass, ECDIS, Radar, BAM, GPS antenna, and VHF antenna. The communication type is serial.

So, we need two things.

1. Your components has been connected to other components or systems.
2. What the communication type is when your components are connected to other components or system.

AIS 시스템을 예시로 확인해보겠습니다.

AIS 시스템은 AIS transponder 와 control panel로 두 가지 구성품을 가집니다.

AIS transponder와 control panel 에서는 LAN 통신을 하게 됩니다.

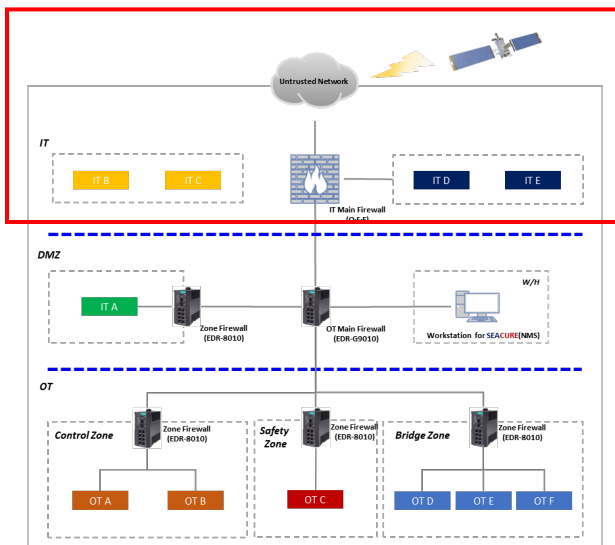
그리고 AIS transponder는 타 시스템과 serial 통신을 하게 됩니다.

이에 귀사의 시스템 각 구성품들이 타 시스템이나 구성품에 물리적 연동이 있는지 확인 부탁드립니다.

또한 해당 연동의 통신 종류가 LAN(IP 통신)인지 Serial 통신인지 알려주시기를 부탁드립니다.

Group 2 – Technical Information (Column P)

Column P Regulation	Column P: (recommended) Connections to Untrusted networks This column should indicate the connections to components and systems that may appear to be not “in the scope of UR E26”. Please note that UR E26 considers any network outside the scope of applicability of UR E26 to be an “Untrusted network”. The intent of this column is to identify possible connections to an untrusted network. In some cases, the subject equipment may not have any connections to an untrusted network. For additional information, please see paragraph 4.2.6 of UR E26 and paragraph 4.2 of UR E27.
Actions taken	1. Please kindly check if your system/components have any connections to any shore systems (Remote maintenance center, SOC, cloud, etc.) and IT systems 2. Please mark Shore if your system/components have any connections to any shore systems. (Please advise us what purpose is.) 3. Please mark IT systems if your system/components have any connections to any IT systems(Smart Ship platform and etc). (Please advise us what purpose is.) 4. Please mark N/A if your system/components don't have any connections to any shore systems and IT systems. 1. 귀사의 시스템이나 구성품이 선박 외부 시스템(원격보수센터, 관제센터, 클라우드 등) 및 IT 시스템에 연결을 가지는 확인 해주십시오. 2. 만약 귀사의 시스템이나 구성품이 선박 외부 시스템에 연결되어 있다면, Shore 로 기재하십시오. (목적도 같이 설명 부탁드립니다.) 3. 만약 귀사의 시스템이나 구성품이 IT 시스템에 연결되어 있다면, IT systems 로 기재하십시오. (목적도 같이 설명 부탁드립니다.) 4. 만약 어느 외부 시스템 및 IT 시스템과도 연결하지 않는다면, N/A로 기재하십시오.



For example, let's check diagram.

“the scope of UR E26” means It covers OT system, not IT systems and shore systems(Check Red Box). UR E26 considers any network outside the scope of applicability of UR E26 as Untrusted network. So, we need to check if your system/components have any connections to an untrusted network.

옆 다이어그램을 예시로 확인해보겠습니다.

IACS UR E26의 시스템 범위는 OT 시스템만을 범위로 하고 있으며, IT 시스템이나 선박 외부 시스템에 대해서는 포함되지 않습니다. 이에 빨간 네모 부분이 Untrusted network로 확인되는 부분입니다.

귀사의 시스템이나 구성품이 선박 외부 시스템이나 IT 시스템에 연결 있는지 확인하고 기재 부탁드립니다.

Group 2 – Technical Information (Column Q, R, S)

Column Q Regulation	Column Q: (paragraph 4.1.1.3.1 in UR E26 requirement) Physical interfaces This column should indicate the physical interfaces for the subject equipment (such as network or serial interfaces). The type and number should be indicated, such as USB*2, RJ45*1, serial, etc.
Actions taken	1. Please kindly identify LAN ports, USB ports, and serial ports on each of your devices. 2. Please mark each port type with number if your devices have any physical interfaces. (Please advise us if the interfaces are in use / in total.) 3. Please separately mark each external drive interface with number if your devices have external drive interfaces (floppy disk, CD-ROM, DVD, SC card etc.) For examples : LAN (1/4), USB (2/4), Serial (1/2), DVD-ROM *1 1. 귀사의 장치마다 보유한 LAN포트, USB포트, Serial포트를 확인해주시요. 2. 귀사의 장치에 물리적 인터페이스가 있을 시, 각 포트 타입 및 수를 기재하십시오. (사용 중인 포트 / 전체 포트를 기재도 같이 부탁드립니다.) 3. 귀사의 장치에 외장디스크(플로피디스크, CD롬, DVD, SD card 등) 인터페이스가 있다면 별도로 기재하십시오.
Column R Regulation	Column R: (paragraph 4.1.1.3.1 in UR E26 requirement) Supported communication protocols This column should indicate the relevant supported communication protocols for the subject equipment. This should include general and private protocols, such as HTTP, SSH, NMEA 0183, MODBUS-TCP, etc.
Actions taken	1. Please mark the protocol type (data format) used by your equipment. It shall be listed all protocols used to communicate with other systems/devices. For examples : HTTPS, MQTT, NMEA 0183, MODBUS-RTU 1. 귀사의 장비에서 이용하는 프로토콜 종류(데이터 포맷)를 기재하십시오. 타 시스템/장비와 통신하는 모든 프로토콜이 기재되어야 합니다.
Column S Regulation	Column S: Recommended (Optional – only if permitted by the Owner’s cyber-security policy) IP ranges/MAC addresses of nodes connected (if applicable) This column should indicate the IP ranges and MAC addresses of the nodes connected (if applicable).
Actions taken	1. Please mark the IP ranges and MAC address of the nodes connected if your system have default values for IP ranges and MAC address. 2. Please remain blank if you don’t want to open IP ranges/MAC addresses due to your cybersecurity policy. 1. 귀사의 시스템에 IP 및 Mac 주소에 대해 Default 값이 있다면, IP 및 MAC 주소를 기재해주시요. 2. 만약 귀사 사이버보안 정책 에 따라 알려지기 원하지 않는다면, 공란으로 나두십시오.



Group 3 – Cybersecurity Information

This group is for cybersecurity information. So, you need to understand the regulation first and then you can fill out the contents in Group 3. But If you don't know, please mark **SeaNet**.
그룹 3은 사이버보안 관련 정보 기재입니다. 이에 관련 규정 이해 먼저하시고 작성 부탁드립니다. 만약, 이해가 안된다고 생각되신다면, **씨넷**으로 기재 부탁드립니다.

Table 3. Examples					
Column E:	Column M:	Column N:	Column T: (Recommended)	Column U: (Recommended)	Column V:
System Category	Security zone	Short description of functionality/purpose	Certificate (Type/ Ship-specific/etc.)	Excluded (UR E26 Sec.6)	Update log (Check the change records to find details)
Cat III	Steering Zone	Steering control system providing control of the rudder	Product-by-Product approval		No update
Cat III	Propulsion Zone	Control system functions and integrate their subsystems.	24CYxxx		No update
Cat III	Propulsion Zone	Control panel for main engine which is located in the wheelhouse	24CYxxx		No update
Cat III	Propulsion Zone	Control panel for main engine which is located in the control room	24CYxxx		2024/08/26 10:05 (UTC) - update software to version 1.05 2024/09/26 12:35 (UTC) - update software to version 1.10
Cat III	Propulsion Zone	Managed ethernet switch which provide external and internal IP connections	24CYxxx		No update
Cat III	Propulsion Zone	Firewall manages ethernet switch, controls and monitors incoming and outgoing network traffic based on predetermined security rules.	24CYxxx		No update
Cat III	Propulsion Zone	Integrates remote control systems	23CYxxx		No update
Cat III	Propulsion Zone	Bridge control functions	23CYxxx		No update
Cat III	Propulsion Zone	Engine Control Room control functions	23CYxxx		No update
Cat III	Propulsion Zone	Emergency control functions	23CYxxx		No update
Cat III	Propulsion Zone	Connect overall systems and the external networks, additionally manages serial Networks	23CYxxx		No update
Cat II	Propulsion Zone	Integrate information from sensors and provide alarm when they sense any events at the intended high level.	22CYxxx		No update
Cat II	Propulsion Zone	Water level sensor - air purge type	-		No update
N/A	Navigation zone	Determine north direction and accurate heading	ABS certificate no.; DNV certificate no.; CCS certificate no.; NK certificate no.		No update
N/A	Navigation zone	Implement of data collection and visualisation, include XX, XX,...(data)	Ship specific certificate		2024/07/09 09:00 (UTC) - update application software to V2.0 (check the change records for details)
Cat III	External Connection Zone	Ship network access control and malware protection	25CYxxx		2024/07/01 09:00 (UTC) - update virus database 2024/07/09 08:00 (UTC) - update the software to v1.1

Group 3 – Cybersecurity Information (Column E)

Column E Regulation	Column E: (paragraph 4.1.1.3.1 in UR E26 requirement) System Category This column should indicate the System Category defined in UR E22 3.1. Most will be either Category II or Category III. Some will be Category I. It is possible the “not applicable” could be entered in this column for equipment that does not fit into one of the three system categories defined in UR E22 3.1. One example would be navigation equipment which is out of scope of UR E22. However, navigation equipment is within the scope of UR E26 and therefore should be in the inventory.
Actions taken	1. Please kindly check what category your system is involved in. For your easy understanding, please kindly find the category definition. 2. Please mark Category I if your system has System functionality as Monitoring, informational and administrative functions. 3. Please mark Category II if your system has System functionality as Vessel alarm, monitoring and control functions which are necessary to maintain the vessel in its normal operational and habitable conditions. 4. Please mark Category III if your system has System functionality as Control functions for maintaining the vessel's propulsion and steering and Vessel safety functions. 1. 귀사의 시스템이 어느 카테고리에 속하는지 확인해주시요. 쉬운 이해를 위해 카테고리 정의를 참고하십시오. 2. 귀사의 시스템이 모니터링, 정보 및 행정 기능만 가지고 있다면, 카테고리 I 으로 기재하십시오. 3. 귀사의 시스템이 선박의 정상적인 운항 및 거주 상태로 유지하기 위해 필요한 선박 알람, 모니터링 및 제어 기능을 가지고 있다면, 카테고리 II 로 기재하십시오. 4. 귀사의 시스템이 선박의 추진 및 조타를 유지하기 위한 제어 기능이나 선박의 안전 기능을 가지고 있다면, 카테고리 III 으로 기재하십시오.

3.1 System category definitions (Reference : IACS UR E22 Rev.3)

The categorization of a system in the context of this UR is based on **the potential severity of the consequences** if the system serving the function fails. Table 3 provides the definitions of the categories.

Category	Failure effects	Typical System functionality
CAT. I	Those systems, failure of which will not lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.	• Monitoring, informational and administrative functions
CAT. II	Those systems, failure of which could eventually lead to dangerous situations for human safety, safety of the vessel and/or threat to the environment.	• Vessel alarm, monitoring and control functions which are necessary to maintain the vessel in its normal operational and habitable conditions
CAT. III	Those systems, failure of which could immediately lead to dangerous or catastrophic situations for human safety, safety of the vessel and/or threat to the environment.	• Control functions for maintaining the vessel's propulsion and steering • Vessel safety functions

3.2 Class Societies' scope

Category I systems are normally not subject to verification by the Class society, as failure of these systems shall not lead to dangerous situations. However, information pertinent to category I systems shall be required upon request to determine the correct category or ensure that they do not influence the operation of systems in category II and category III.

Group 3 – Cybersecurity Information (Column E)

3.3 System category examples

The category of a system shall always be evaluated in the context of the specific vessel in question; thus, the categorization of a system may vary from one vessel to the next. This means that the examples of categories below are given as guidance only. For determining the categorization of systems for a specific vessel, see paragraph 4.3.3.

Category (IACS UR E22)	Function (IACS UR E22)	Systems (IACS UR E22)	Systems (IACS Rec 171)	Detailed Example of Systems (IACS Rec 171)
CAT. I	Monitoring, informational and administrative functions	- Fuel monitoring system, - maintenance support system, - diagnostics and troubleshooting system, - closed circuit television, - cabin security, - entertainment system, - fish detection system		- Satcom/VSAT - Satcom/FBB - VDR - Weather Fax - Anemometer - Incinerator
CAT. II	Vessel alarm, monitoring and control functions which are necessary to maintain the vessel in its normal operational and habitable conditions	- Fuel oil treatment system, - alarm monitoring and safety systems for propulsion and auxiliary machinery, - Inert gas system, - control, - monitoring and safety system for cargo containment system.	- Liquid cargo transfer control system, - Bilge level detection and associated control of pumps, - Fuel oil treatment system, - Ballast transfer valve remote control system, - Stabilization and ride control systems, - Alarm and monitoring systems for propulsion systems	- Boiler Management system - Dynamic positioning system - IAS(Integrated Automation system; when IAS only collects information) - Ballast water management system - MF/HF - Inmarsat-C - Navtex Receiver - Speed Log - AIS - BNWAS - Sewage treatment system
CAT. III	- Control functions for maintaining the vessel's propulsion and steering - Vessel safety functions	- Propulsion control system, - steering gear control system, - electric power system (including power management system), - dynamic positioning system (IMO classes 2 and 3).	- Propulsion system of a ship, meaning the means to generate and control mechanical thrust in order to move the ship (devices used only during manoeuvring are not in the scope of this requirement such as bow tunnel thrusters), - Steering system control system, - Electric power system (including power management system), - Ship safety systems covering fire detection and fighting, flooding detection and fighting, internal communication systems involved in evacuation phases, ship systems involved in the operation of life saving appliances equipment, - Dynamic positioning system of equipment classes 2 and 3 according to IMO MSC/Circ.645.	- Main Engine - Remote Propulsion control system - Steering Gear system - Auxiliary Engine - Dynamic positioning system - Emergency generator control and safety system - IAS(Integrated Automation system; when IAS interacts with the systems to which it is connected) - Remote Tank gauging system - Cargo Control and monitoring system(loading system) - Oil discharge monitoring and control system(ODME) - Inert Gas system - Loading computer - VHF - Gyro compass - ECDIS - Echo Sounder - GNSS receiver - RADAR - Heading or track control system - Fire detection system - Gas detection/Gas Sampling system - Emergency shutdown system

Group 3 – Cybersecurity Information (Column M, N, T)

Column M Regulation	Column M: (paragraph 4.1.1.3.1 in UR E26 requirement) Security zone This column should identify the security zone that the subject equipment belongs to.
Actions taken	1. Please kindly check what function your system is involved in. For your understanding, please find the function in system examples in Column B. 2. Please mark the relevant function security zone such as Propulsion zone, steering zone, safety systems zone, navigation zone, external communication zone and other zone. 3. Please mark SeaNet if you don't know what the security zone your system belongs to. 1. 귀사의 시스템이 어느 기능을 하고 있는지 확인하십시오. Column B에 시스템 예제 상에 기능들을 보시면 됩니다. 2. 관련 기능 보안 영역 을 기재해주십시오. (예, Propulsion zone ~ other zone) 3. 만약 귀사의 시스템이 어느 보안 영역인지 알 수 없다면, SeaNet 으로 기재 부탁드립니다.
Column N Regulation	Column N: (paragraph 4.1.1.3.1 in UR E26 requirement) Short description of functionality/purpose This column should indicate a short description of the functionality/purpose . This information is needed for hardware and software items.
Actions taken	1. Please kindly mark a shore description of the functionality and purpose about your system/equipment. 2. Please kindly mark a shore description of the functionality and purpose about your software. 1. 귀사 시스템/장치 기능 및 목적에 대한 짧은 설명을 기재해주십시오. 2. 귀사의 소프트웨어 의 기능 및 목적에 대한 짧은 설명을 기재해주십시오.
Column T Regulation	Column T: (recommended) Certificate: Type approval, ship-specific approval, etc. (if applicable) This column should indicate information concerning any approval by a Class Society , such as a type approval, ship-specific approval, etc. (if applicable). The Class Society and certificate number should be indicated for the system or component that has been issued an approval covering the requirements in UR E26/E27.
Actions taken	1. Please kindly mark certificate number if you have a cybersecurity type approval about your system/equipment issued by the relevant class society. 2. Please kindly mark In progress if your system/equipment is under consideration from the relevant class society. Please inform us when the certificate is expected to be issued. 3. Please kindly mark N/A if you don't have the type approval about your system/equipment. Please advise us if you have any intention to have the type approval or exemption cert. 1. 관련 선급 으로부터 귀사의 시스템이나 장치가 사이버보안 type approval 를 가지고 있다면, certificate number 를 기재해주십시오. 2. Type approval 진행 중이라면, In progress 로 기재해주십시오. 또한 언제쯤 증서를 수령하실 수 있는지 예상 일정 알려주십시오. 3. Type approval 가 없을 경우, N/A 기재해주십시오. 그리고 Type approval 을 받거나 면제 증서를 받을 계획이 있다면 알려 주십시오.

Group 3 – Cybersecurity Information (Column U & V)

Column U Regulation	Column U: (recommended) Excluded (If part of an excluded system in accordance with UR E26, Section 6) This column should remain blank for most equipment. This column should indicate an “X” if the equipment is part of a computer-based system (CBS) that has been excluded from meeting the requirements of UR E26. See Section 6 of UR E26, which would include submission of the “Risk assessment for exclusion of CBS from application of requirements” which would be subject to review by the Class Society.
Actions taken	1. Please kindly remain blank. System Integrator(SeaNet) will carry out the risk assessment for exclusion of CBS and mark X if your system is excluded from the target system. 1. 공란으로 남겨주십시오. 시스템 통합자가 CBS 제외를 위한 위험 평가를 시행하여, 대상 시스템에서 제외된다면 X로 표시할 것입니다.

Column V Regulation	Column V: (paragraph 4.1.1.3 in UR E26 requirement) Update log This column should indicate an update log for the subject items. Software and hardware modifications potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems should be indicated in the update log column. Information should indicate when the modification was made, what the modification was and provide direction to Change Records.
Actions taken	1. Please kindly check if your software and hardware have any updates(modifications). 2. Please kindly mark when the modifications happened and what it was about your software and hardware modification if you have any updates(modification). 3. Please kindly mark No update if your system/equipment doesn't have any updates (modifications). 1. 귀사 소프트웨어 및 하드웨어에 업데이트(수정사항)이 있는지 확인해주십시오. 2. 업데이트(수정사항)이 있다면, 언제 수정이 발생했고 어떤 부분을 수정하였는지 기재해주십시오. 3. 업데이트(수정사항)이 없다면, No update로 기재해주십시오.



Sea Earth Air Network

We are building a net with sea, earth, and air



Homepage

<http://www.sea-net.co.kr/>



Facebook

<https://www.facebook.com/SeaNet.korea/>



Buykorea

www.buykorea.org (Please search the 'SeaNet')

Busan Headquarters Tel.	051-417-5802
Seoul Office Tel.	02-6952-5752
Geoje Office Tel.	055-682-6934
Ulsan Office Tel.	051-405-0746
Email.	CS@sea-net.co.kr
